

PathFinder: Negotiated-Congestion Rip-up-and-Reroute for Minor Embedding on D-Wave Quantum Annealers

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Minor embedding—mapping a logical problem graph onto the fixed sparse connectivity of a quantum annealer by representing each logical variable as a connected *chain* of physical qubits—is the compilation bottleneck for quantum annealing. The de facto standard, *minorminer*, is a randomized rip-up-and-reroute heuristic whose chains are built as unions of independent shortest paths and whose congestion signal is a memoryless per-pass snapshot; both choices inflate the *average chain length* (ACL), the metric most widely taken as the primary proxy for embedded-problem solution quality, and produce high run-to-run variance. We observe that chain construction is exactly the multi-net global routing problem solved for thirty years in FPGA/VLSI computer-aided design, and port its two key ideas to minor embedding: a true Steiner-tree inner step (the shortest-path heuristic) in place of independent shortest paths, and *negotiated congestion* (McMurchie–Ebeling) in place of the snapshot penalty. We package these as PathFinder, an anytime, large-neighbourhood *improver* that warm-starts from a fast base embedding and then rips up and reroutes the longest chains through a congestion-priced fabric, accepting a move only if it lowers the total qubit count and preserves validity—so it never returns an embedding worse than its base. The algorithmic primitives are classical; the contribution is their synthesis and application to minor embedding, together with an empirical evaluation inside the Ember benchmarking framework. On Erdős–Rényi, d -regular, and Barabási–Albert sources embedded into clean and broken D-Wave Pegasus and into Zephyr (all four methods reach 100% success on every target), pathfinder never returns a longer-ACL embedding than minorminer on any of the 35 instance–topology cells, and its thorough configuration lowers mean ACL on every cell (by 1–11%, mean $\sim 5\%$) and reduces ACL standard deviation in 30 of the 35 (by up to 77%), at the cost of additional wall-clock time. We then optimize the algorithm—bounded-region routing, a dirty-set incremental schedule, and a spur-pruning finishing pass—making the production improver $\sim 3\times$ faster at equal-or-better ACL: single-seed pathfinder runs within $\sim 1.3\times$ compiled minorminer’s wall-clock while still shortening mean ACL, and the optimized thorough configuration reaches -5% mean ACL (up to -11%) at a fraction of its former cost.

CCS Concepts: • **Hardware** $\rightarrow$ **Quantum computation**; • **Theory of computation** $\rightarrow$ *Routing and layout*; • **Mathematics of computing** $\rightarrow$ *Graph algorithms*.

Additional Key Words and Phrases: minor embedding, quantum annealing, D-Wave, graph minors, negotiated congestion routing, Steiner tree, large-neighbourhood search

1 INTRODUCTION

Quantum annealers such as those built by D-Wave solve optimization problems encoded as Ising/QUBO models whose interaction graph is the user’s *logical* (problem) graph. The hardware, however, exposes only a fixed, sparse *physical* graph—successive D-Wave generations Chimera, Pegasus [2], and Zephyr [3] have qubit degree 6, 15, and 20 respectively. Running a problem whose interaction graph is denser than the hardware requires *minor embedding*: each logical variable is realized by a connected set of physical qubits, a *chain*, whose qubits are forced into agreement by strong ferromagnetic couplings so that the chain behaves as one logical spin [5, 9].

The minor-embedding problem. Let H be the *source* (problem) graph and G the *target* (hardware) graph. A minor embedding is a map $\varphi : V(H) \rightarrow 2^{V(G)}$ assigning to each logical vertex v a chain $\varphi(v) \subseteq V(G)$, subject to three conditions:

- (1) **Connectivity.** Each $\varphi(v)$ induces a connected subgraph of G .
- (2) **Disjointness.** The chains are pairwise disjoint: $\varphi(u) \cap \varphi(v) = \emptyset$ for $u \neq v$.
- (3) **Edge preservation.** For every edge $(u, v) \in E(H)$ there is at least one hardware edge between the chains, i.e. some $a \in \varphi(u)$, $b \in \varphi(v)$ with $(a, b) \in E(G)$.

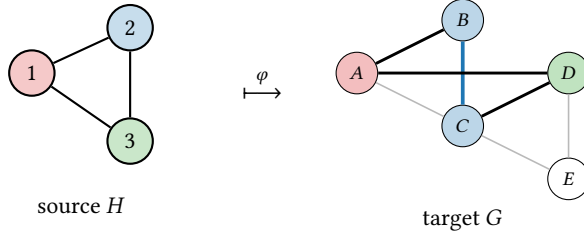


Fig. 1. Minor embedding of the triangle H into a hardware patch G . Vertex $1 \mapsto \varphi(1) = \{A\}$, $2 \mapsto \{B, C\}$, $3 \mapsto \{D\}$ (matching colours). Chain $\varphi(2)$ is connected through the bold intra-chain coupler $B-C$; the three chains are disjoint; each H -edge is realized by a hardware coupler between the corresponding chains (1–2 by $A-B$, 2–3 by $C-D$, 1–3 by $A-D$). Qubit E is unused. Here $ACL = \frac{1+2+1}{3} = \frac{4}{3}$.

The map is total with non-empty chains, so every logical vertex is realized; Ember’s validator (§4) records this as the *coverage* check. Figure 1 illustrates the conditions. Feasibility is the question “does a minor embedding exist?”; in practice one wants the *best* embedding, because the dominant cost is qubit usage and chain fidelity. The standard objective is to minimize the *average chain length*

$$ACL(\varphi) = \frac{1}{|V(H)|} \sum_{v \in V(H)} |\varphi(v)|, \quad (1)$$

and to keep chain lengths *uniform* (low coefficient of variation): long and unequal chains require stronger chain couplings, shrink the available dynamic range for the logical problem, and empirically raise the annealer’s solution error. ACL is therefore the headline quality metric, with total qubit count, maximum chain length, and chain-length uniformity as close companions.

Contribution. This paper makes four contributions. (i) We frame minor-embedding chain construction as *multi-net global routing* and transfer two ideas from FPGA/VLSI computer-aided design—a real Steiner-tree inner step and *negotiated congestion*—to the quantum-annealing setting (§3). (ii) We package them as **PathFinder**, an anytime, large-neighbourhood *improver* that warm-starts from a fast base embedding and, by construction, never regresses below it. (iii) We evaluate PathFinder inside the Ember benchmarking framework (§4) against minorminer and minorminer-with-p-norm-layout on clean and broken Pegasus and on Zephyr (§5), showing it reduces mean ACL and—the property that most distinguishes it—run-to-run ACL *variance* in nearly every cell, while all methods retain full success and validity. (iv) We then *optimize* the algorithm (§6): a structured pass yields three quality-safe improvements—bounded-region routing, a dirty-set incremental schedule, and a spur-pruning finishing pass—that together make the production algorithm $\sim 3\times$ faster at equal-or-better ACL, bringing single-seed pathfinder to within $\sim 1.3\times$ compiled minorminer’s wall-clock. We are explicit (§3.5) that the algorithmic primitives are not new; the novelty is their synthesis and application to minor embedding.

2 BACKGROUND AND RELATED WORK

The standard heuristic and its weaknesses. minorminer (MM), the Cai–Maready–Roy heuristic [5], is randomized coordinate descent. It fixes a random vertex order and, for each vertex, rips out its chain and rebuilds it as the union of weighted shortest-path computations to its already-placed neighbours; qubit contention is discouraged by a vertex weight $\text{diam}(G)^k$ where k is the number of chains currently using the qubit, recomputed from scratch each pass; it iterates until the embedding is valid or a patience bound is hit. Three structural weaknesses follow directly. First, a chain is a *union of independent shortest paths*—a weak Steiner-tree heuristic that misses shared structure.

Second, the congestion signal is *myopic*: a per-pass snapshot with no memory of how contested a qubit has been over time. Third, the result is highly *order- and seed-dependent*. The most recent state-of-the-art evaluation [7] documents the consequences on broken Pegasus: run-to-run ACL can differ by several qubits per chain on the same instance, MM is beaten by deterministic clique embedding once the source’s average degree exceeds the hardware degree, and MM routinely hits a ~ 1000 s cutoff on large dense instances. The MM paper itself names “better initial placement of vertex-models” as the key open problem.

Deterministic templates. For sufficiently structured inputs, embeddings can be read off a template. Clique (“busclique”) embedding generates complete-graph minors in Chimera and Pegasus connectivity directly [4]; complete-bipartite templates [16] and the 4-clique network on a contracted hardware graph [14] extend the idea. These are fast and uniform but over-provision qubits on graphs that are not near-complete.

Layout- and spring-seeded heuristics. A complementary line attacks MM’s initial-placement weakness by first laying out the source graph in a low-dimensional space and placing variables near the geometrically closest qubits: layout-aware embedding [15], the spring/clique-seeded methods of Zbinden et al. [21], and Ocean’s “Layout Embedding”, which seeds MM with a p -norm layout. We use the latter as a baseline.

Exact methods. Integer-programming formulations with Benders decomposition [1] and algebraic-geometry/Gröbner-basis compilers [6] solve minor embedding optimally but do not scale past tens of vertices.

Metaheuristics and learned policies. Path/super-vertex simulated annealing (PSSA) [17] and graph-theoretic OCT/virtual-hardware methods [8] apply local search; more recently, GNN-RL (CHARME [13]) and PPO policies [12] learn chain-construction heuristics.

Routing and optimal transport. Two threads inform our approach. Negotiated-congestion routing (PathFinder) [10] from FPGA CAD treats each net as a resource consumer that negotiates for congested wires over iterations; and Gromov–Wasserstein optimal transport [19, 20] matches graphs by their intra-graph distances, suggesting a global, continuous view of the assignment. Our method develops the first thread; the second is promising future work (§7).

3 THE PRELIMINARY PATHFINDER ALGORITHM

This section presents PathFinder as first designed; §5 reports its results. A subsequent optimization pass (§6) makes the production algorithm $\sim 3\times$ faster at equal-or-better quality without changing the ideas below, so we keep this preliminary description intact and report the optimized numbers separately.

3.1 Minor embedding as multi-net routing

A chain $\phi(v)$ that must be connected and adjacent to each neighbour’s chain is exactly a *Steiner tree* (a routing “net”) whose terminals are the qubits bordering the neighbour chains, and the disjointness requirement is exactly the rule that two nets may not share a routing resource. Minor embedding is therefore a multi-net global routing problem on the graph G , and the mature machinery of FPGA/VLSI routing applies. We import its two central ideas.

3.2 A real Steiner inner step

Rather than build a chain as a union of independent shortest paths to a single root (MM), we grow one tree with the classical *shortest-path heuristic* (SPH) [11, 18]: starting from a root, each

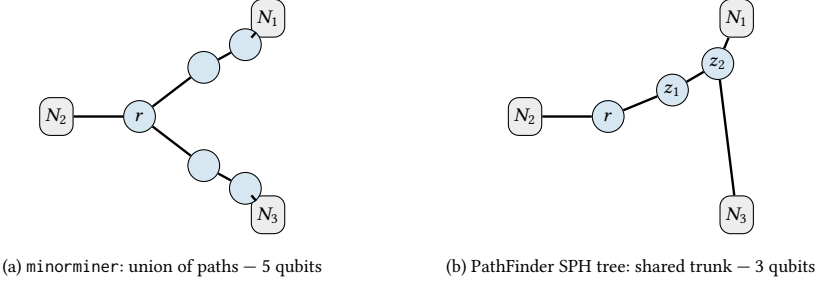


Fig. 2. Inner step contrast for a chain that must reach three neighbour chains N_1, N_2, N_3 . (a) MM routes an independent shortest path to each, duplicating qubits when N_1 and N_3 are near each other. (b) The shortest-path-heuristic Steiner tree attaches N_3 to the nearest existing tree node (z_2), reusing the trunk built for N_1 , yielding a shorter chain.

neighbour terminal is connected to the *nearest node of the current tree*, so later connections reuse earlier structure. Figure 2 contrasts the two: when neighbours cluster, SPH shares a trunk and produces a strictly shorter chain.

3.3 Negotiated congestion

We replace MM’s snapshot penalty with the negotiated-congestion cost of McMurchie and Ebeling [10]. Routing prices each qubit q at

$$\text{cost}(q) = (1 + h_q) (1 + \lambda \cdot o_q), \quad (2)$$

where o_q is the number of *other* chains currently occupying q (occupancy), λ is the present-congestion factor, and h_q is a *history* term that *accumulates* by a fixed increment every iteration in which q is over-subscribed. The present term makes a qubit more expensive as nets pile onto it within a pass; the history term gives the price a memory, so a persistently contested qubit becomes permanently expensive and the chains that can most cheaply route around it eventually yield. This is a Lagrangian view of the disjointness constraint; in FPGA routing it is known to drive the search toward a legal, low-congestion state, and we observe the same behaviour here—behaviour MM’s memoryless snapshot cannot sustain. A node-weighted Dijkstra over Eq. (2) supplies the shortest-path computations the SPH step needs.

3.4 A warm-started large-neighbourhood improver

A from-scratch router that places every chain from scattered single-qubit seeds is throttled by exactly the open problem MM names—*initial placement*. Without co-locating adjacent variables, dense-graph chains balloon and the search often fails to legalize at all. The standalone pathfinder-cold variant (a cold start followed by the same improver) bears this out: on the nine clean-Pegasus Erdős-Rényi instances of our grid ($n \in \{20, 30, 40\}$, $d \in \{0.3, 0.5, 0.7\}$) it legalizes only four—the smaller and sparser ones, at ACL 2.5–3.8—and fails on the five larger, denser ones, whereas minorminer embeds all nine (reproduced by docs/paper/data/coldstart_probe.py). PathFinder therefore runs as an *improver*. It warm-starts from a fast base embedding (MM by default; any registered algorithm works), then performs *large-neighbourhood* rip-up-and-reroute—the neighbourhood being a long chain together with the cascade of chains its rerouting displaces. It takes the longest chain and lets it re-route along the cheapest tree under Eq. (2)—which may cross occupied territory and create temporary overlaps—then re-routes every chain it displaced through free space. A move is committed only if it lowers the embedding’s total qubit count (equivalently, its ACL) and leaves

Algorithm 1 PathFinder (driver)**Require:** source H , target G , base embedder $\mathcal{B}$

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1:  $\Phi \leftarrow \mathcal{B}(H, G)$  ▷ warm start
2: if  $\Phi$  invalid then  $\Phi \leftarrow \text{COLDSTART}(H, G)$  ▷ BFS-ordered negotiated fallback
3: end if
4: if  $\Phi$  invalid then return failure
5: end if
6:  $\Phi^* \leftarrow \Phi$  ▷ best valid so far
7: repeat
8:   for all  $v \in V(H)$  in decreasing chain length do
9:      $\Phi \leftarrow \text{TRYSHORTEN}(\Phi, v)$  ▷ Alg. 2
10:    if  $\text{ACL}(\Phi) < \text{ACL}(\Phi^*)$  then  $\Phi^* \leftarrow \Phi$ 
11:    end if
12:  end for
13: until no improvement or deadline
14: return  $\Phi^*$ 

```

Algorithm 2 TRYSHORTEN(Φ, v) — one LNS destroy-repair move

```

1: snapshot  $\Phi$ 
2: rip up  $\varphi(v)$ ; rebuild it as the cheapest SPH tree under Eq. (2) ▷ may overlap others
3: if new chain not shorter then restore; return  $\Phi$ 
4: end if
5: for all chain  $w$  now overlapping  $\varphi(v)$  do
6:   rip up  $\varphi(w)$ ; reroute it through free qubits only
7:   if no free-space route then restore; return  $\Phi$ 
8:   end if
9: end for
10: if total qubits decreased and  $\Phi$  valid then return  $\Phi$ 
11: else restore; return  $\Phi$ 
12: end if

```

it valid; otherwise the prior state is restored (Algorithms 1–2). Because PathFinder keeps the best valid embedding it has seen, it **never returns an embedding worse than its base**, and it spends its entire time budget reducing ACL—an anytime guarantee MM’s one-shot, patience-bounded run lacks. The thorough configuration additionally runs four seeded restarts and keeps the best, trading wall-clock for lower ACL and variance. When no valid base is available, a fallback builds the initial embedding from scratch in breadth-first vertex order under the same negotiated cost; uncompetitive on its own, it exists only to guarantee an output.

3.5 Novelty

We state plainly what is and is not new. Negotiated-congestion routing is due to McMurchie and Ebeling [10]; the shortest-path Steiner heuristic is classical [11, 18]; and large-neighbourhood/destroy-repair search is a standard matheuristic paradigm. *None of these primitives is our invention.* The contribution is their synthesis and their application to quantum-annealing minor embedding: casting chain construction as multi-net routing, replacing MM’s two weakest components (its inner step and its congestion signal) in a single drop-in algorithm, running it as a warm-started

anytime improver with a never-regress guarantee, and validating empirically that this reduces ACL and—more importantly—ACL variance against the standard tool. The clean substitution also isolates a question of independent interest: *is it MM’s congestion signal or its inner step that costs the most?*

4 EMBER AND EVALUATION METHODOLOGY

We evaluate inside **Ember** (ember-qc), an open benchmarking framework for minor-embedding algorithms on D-Wave topologies. Every algorithm implements a common interface and is run through Ember’s atomic harness `benchmark_one`, which times the call and computes quality metrics (chain lengths, ACL, max chain length, qubits, couplers). Before any metric is trusted, the output passes four-layer validation: type and format checks followed by the structural checks of §1 (coverage, connectivity, disjointness, edge preservation). Targets are generated with `dwave-networkx` and may have qubits removed to simulate fabrication faults, and per-trial seeds are derived deterministically by SHA-256 of the task identity, so runs are reproducible and order-independent. Results aggregate into per-group means and standard deviations.

Protocol. Targets are clean Pegasus P_6 , a *broken* Pegasus P_6 with 5% of qubits removed at random (the state-of-the-art evaluation setting), and Zephyr Z_4 . Sources are an Erdős–Rényi (ER) grid over $n \in \{20, 30, 40\}$ and edge density $d \in \{0.3, 0.5, 0.7\}$, plus d -regular and Barabási–Albert graphs for generality; one fixed instance per cell, so the reported spread is the run-to-run variance *on a fixed instance*—MM’s documented failure mode. Baselines are `minorminer` and `minorminer-layout` (p -norm layout). Each (algorithm, instance) is run over five seeds; we report success rate, ACL mean and standard deviation, max chain length, qubits, and wall-clock time.

5 PRELIMINARY RESULTS

These are the results of the preliminary algorithm of §3; the optimized production version and its (improved, and much faster) numbers follow in §6. All four methods embed every instance across all seeds and topologies (100% success), so success is not a differentiator on this grid; the comparison is on chain length, its variance, and time. Table 1 reports embedding quality on clean Pegasus P_6 , Table 2 the corresponding wall-clock time, and Table 3 robustness across topologies; Figure 3 shows the density trend.

Two findings are sharp. First, the single-seed pathfinder *never regresses*: across all 35 instance-topology cells (every per-seed run, in fact) its mean ACL is $\leq$ `minorminer`’s—the maximum observed difference is 0.000 to three decimals—confirming the best-valid guarantee empirically. Second, `pathfinder-thorough` lowers mean ACL on the six shown cells by 3.2–4.5%, and reduces ACL standard deviation in five of the six—by up to 76% on the full clean- P_6 grid (e.g. $0.11 \rightarrow 0.03$ at $n=30, d=0.3$). The lone clean- P_6 exception is $n=40, d=0.3$, where `minorminer`’s spread is already at the floor (0.05); we report it rather than hide it. The complete 35-cell grid across all three source families is in Appendix A: there `pathfinder-thorough` lowers mean ACL on *every* cell (by 1–11%; the largest single gain, –10.6%, is on a d -regular source) and cuts ACL standard deviation in 30 of the 35.

Discussion. The single-seed pathfinder is the fair *equal-base* comparison: it consumes the same `minorminer` base for a given seed and only refines it, so it cannot regress—and it already trims ACL on denser cells while leaving the sparse and d -regular cells (where the base is near-optimal) unchanged. The substantive gains come from `pathfinder-thorough`, whose four restarts both lower the mean and, by selecting the best of several runs, sharply cut the run-to-run variance that is `minorminer`’s signature weakness. The effect persists on broken Pegasus—where `minorminer`’s ACL

Table 1. Embedding quality on clean Pegasus P_6 for ER sources (mean over 5 seeds; ACL std in parentheses; lower ACL is better). Best ACL per cell in bold.

cell	algorithm	ACL (std)	max	qubits
$n=30, d=0.3$	minorminer	2.70 (0.11)	3.8	81.0
	minorminer-layout	2.75 (0.15)	4.0	82.4
	pathfinder	2.69 (0.10)	3.8	80.6
	pathfinder-thorough	2.59 (0.03)	3.8	77.6
$n=30, d=0.5$	minorminer	3.30 (0.16)	4.6	99.0
	minorminer-layout	3.43 (0.10)	4.8	102.8
	pathfinder	3.25 (0.15)	4.6	97.6
	pathfinder-thorough	3.19 (0.10)	4.0	95.8
$n=30, d=0.7$	minorminer	3.83 (0.22)	5.0	114.8
	minorminer-layout	3.80 (0.17)	5.4	114.0
	pathfinder	3.79 (0.23)	5.0	113.8
	pathfinder-thorough	3.65 (0.06)	4.8	109.6
$n=40, d=0.3$	minorminer	3.56 (0.05)	5.2	142.4
	minorminer-layout	3.62 (0.14)	5.4	144.8
	pathfinder	3.53 (0.06)	5.2	141.0
	pathfinder-thorough	3.41 (0.06)	5.0	136.4
$n=40, d=0.5$	minorminer	4.57 (0.10)	6.4	182.6
	minorminer-layout	4.72 (0.13)	6.8	188.8
	pathfinder	4.49 (0.14)	6.2	179.6
	pathfinder-thorough	4.39 (0.08)	6.2	175.4
$n=40, d=0.7$	minorminer	5.15 (0.06)	7.2	206.0
	minorminer-layout	5.23 (0.17)	7.0	209.2
	pathfinder	5.07 (0.04)	7.0	202.6
	pathfinder-thorough	4.98 (0.04)	7.0	199.0

Table 2. Wall-clock time (mean seconds per embedding, clean P_6). PathFinder is pure Python; minorminer is compiled C++. pathfinder-thorough runs four seeded restarts and may use up to the full 20 s time budget.

cell	mm	mm-layout	pathf.	pf-thor.
$n=30, d=0.3$	0.33	0.33	1.11	20.08
$n=30, d=0.5$	4.44	3.30	9.04	18.82
$n=30, d=0.7$	1.22	0.98	3.23	12.33
$n=40, d=0.3$	0.54	0.46	2.12	8.37
$n=40, d=0.5$	1.11	0.99	4.12	16.41
$n=40, d=0.7$	1.35	1.73	5.87	20.01

inflates most and the thorough gain is largest—and on Zephyr, so the mechanism is not topology-specific. The cost is wall-clock time: PathFinder layers a pure-Python router on top of the base, and pathfinder-thorough, with its four restarts, can use the full 20 s budget; minorminer-layout, by contrast, is neither consistently faster nor consistently better than plain minorminer.

Table 3. Robustness across topologies for the $n=40, d=0.5$ ER source: ACL (std) and time. minorminer’s ACL inflates most on broken Pegasus, where pathfinder–thorough’s relative gain is largest (-7.3%).

target	algorithm	ACL (std)	time (s)
Pegasus P_6	minorminer	4.57 (0.10)	1.11
	minorminer–layout	4.72 (0.13)	0.99
	pathfinder	4.49 (0.14)	4.12
	pathfinder–thorough	4.39 (0.08)	16.41
broken P_6 (5%)	minorminer	4.88 (0.18)	0.82
	minorminer–layout	4.72 (0.19)	1.19
	pathfinder	4.80 (0.18)	3.52
	pathfinder–thorough	4.52 (0.06)	14.26
Zephyr Z_4	minorminer	3.63 (0.08)	0.98
	minorminer–layout	3.61 (0.16)	1.10
	pathfinder	3.57 (0.07)	3.81
	pathfinder–thorough	3.52 (0.03)	13.90

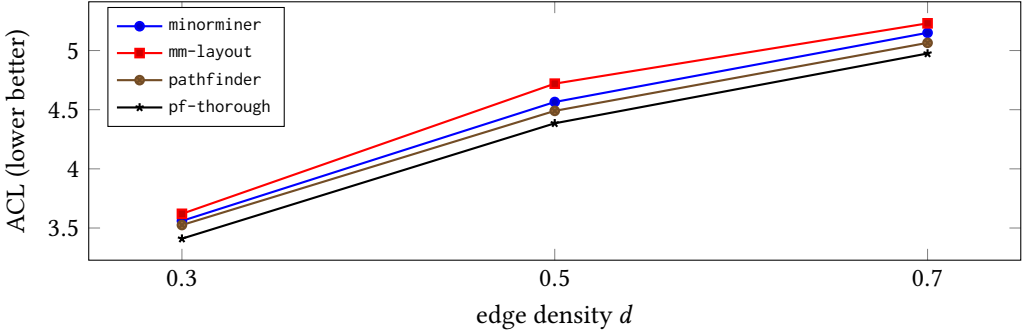


Fig. 3. ACL versus edge density for $n=40$ ER sources on clean Pegasus P_6 . pathfinder–thorough (bottom curve) is uniformly shortest; the gap to minorminer is roughly constant in absolute terms across densities.

6 OPTIMIZATIONS

The preliminary algorithm wins on chain length and variance but pays in wall-clock: its pure-Python router is $\sim 3\text{--}6\times$ slower than compiled minorminer (Table 2). We therefore ran a structured optimization pass—eight candidate improvements (five targeting speed, three quality), each implemented as an isolated variant and measured against the frozen preliminary algorithm. Three quality-safe winners survived and are baked into the production algorithm; the rest were rejected or deferred.

What we bake in. The three production optimizations are mutually orthogonal (each rewrites a different stage of the router) and compose without interaction:

- **Bounded-region routing.** Each per-neighbour shortest-path search is restricted to a breadth-first ball around the vertex’s current chain and its placed-neighbour chains, and stops as soon as the neighbour boundary is settled, instead of computing a full single-source shortest-path tree over the whole 680–5640-qubit fabric. Alone this is $\sim 3\times$ faster with byte-identical chains.

- **Dirty-set incremental scheduling.** The improvement loop keeps a worklist of vertices whose neighbourhood changed rather than re-sweeping every chain each round; it performs the same accepted moves with fewer wasted re-examinations.
- **Spur-pruning.** A deterministic finishing pass removes redundant chain leaves—qubits whose removal keeps the chain connected and breaks no incident edge—to a fixpoint. This is the cheap shadow of exact region repair: on minorminer’s own output it recovers about half of a CP-SAT repair’s ACL gain at $\sim 5000\times$ less time, and on PathFinder it shaves a further $\sim 0.6\%$ ACL for free.

A fourth idea—seeding the improver from a multilevel placement instead of minorminer—lowers variance on dense Pegasus, but the multilevel base is unreliable on small, sparse, and Zephyr instances (it inflated ACL by up to 30% where minorminer is already near-optimal), so we keep it as an optional variant rather than the default. We also rejected an A^* routing scheme (a $+1.7\%$ ACL regression) and an exact Dreyfus–Wagner Steiner inner step (no gain over the shortest-path heuristic), and left a compiled (numba) routing core and parallelized restarts as future speed work.

Optimized results. Tables 4 and 5 report the optimized algorithm on the clean-Pegasus grid. The optimizations are quality-safe: optimized pathfinder is never worse than the preliminary engine and, thanks to spur-pruning, slightly better—mean ACL -1.3% vs. minorminer over all 35 cells (vs. -0.9% for the preliminary engine), and -2.1% vs. minorminer–layout. The headline is speed: single-seed pathfinder now runs in $\sim 1.3\times$ minorminer’s wall-clock (down from $\sim 3\text{--}5\times$) while producing shorter chains, and the optimized pathfinder–thorough delivers -5.1% mean ACL vs. minorminer (up to -10.6% ; -5.9% vs. layout) at roughly the former cost of a *single* preliminary run rather than four. The optimized PathFinder thus keeps the quality advantage of \$5 while closing most of the wall-clock gap to compiled minorminer.

7 LIMITATIONS AND FUTURE WORK

PathFinder’s routing is pure Python. The optimizations of \$6 bring single-seed pathfinder to within $\sim 1.3\times$ minorminer’s wall-clock via bounded-region routing; the remaining gap (and the thorough configuration’s restarts) would close with a compiled (numba) Dijkstra core—which we measured bit-identical and $\sim 2\times$ faster, omitted from the default only to avoid the dependency—and with parallelized restarts. The standalone cold start remains uncompetitive precisely because of the initial-placement problem; pairing PathFinder with a stronger global placement—e.g. a semi-relaxed Gromov–Wasserstein assignment [19]—is the natural next step. The destroy-repair neighbourhood here is a single-chain shortcut; a larger neighbourhood repaired exactly with a constraint solver [1] is a promising route to deeper ACL reductions. Finally, our grid uses one graph instance per (size, density) cell with five seeds; replicating each cell over several instances, with significance tests or confidence intervals, would more cleanly separate algorithm-seed variance from graph-draw variance and further substantiate the variance claims that are PathFinder’s headline advantage.

8 CONCLUSION

Minor-embedding chain construction is multi-net routing, and the routing community has spent decades on exactly the two operations minorminer performs weakly: building a net (Steiner tree) and resolving contention (congestion). Importing a real Steiner inner step and negotiated congestion, and running them as a warm-started anytime improver, yields PathFinder—a drop-in method that never regresses below minorminer and, in its thorough configuration, reduces average chain length and—in nearly every cell—its run-to-run variance on D-Wave Pegasus and Zephyr. A structured optimization pass (bounded-region routing, dirty-set scheduling, spur-pruning) then makes the production algorithm $\sim 3\times$ faster at equal-or-better quality, bringing single-seed pathfinder to

Table 4. **Optimized** PathFinder, embedding quality on clean Pegasus P_6 (ER sources, mean over 5 seeds; ACL std in parentheses; lower is better). Best ACL per cell in bold. Cf. the preliminary Table 1; the optimized pathfinder matches-or-beats it everywhere at $\sim 3\times$ less time (Table 5).

cell	algorithm	ACL (std)	qubits
$n=30, d=0.3$	minorminer	2.70 (0.11)	81.0
	minorminer-layout	2.75 (0.15)	82.4
	pathfinder	2.68 (0.11)	80.4
	pathfinder-thorough	2.58 (0.03)	77.4
$n=30, d=0.5$	minorminer	3.30 (0.16)	99.0
	minorminer-layout	3.43 (0.10)	102.8
	pathfinder	3.23 (0.15)	97.0
	pathfinder-thorough	3.17 (0.11)	95.0
$n=30, d=0.7$	minorminer	3.83 (0.22)	114.8
	minorminer-layout	3.80 (0.17)	114.0
	pathfinder	3.77 (0.20)	113.0
	pathfinder-thorough	3.65 (0.07)	109.4
$n=40, d=0.3$	minorminer	3.56 (0.05)	142.4
	minorminer-layout	3.62 (0.14)	144.8
	pathfinder	3.51 (0.06)	140.4
	pathfinder-thorough	3.39 (0.06)	135.6
$n=40, d=0.5$	minorminer	4.57 (0.10)	182.6
	minorminer-layout	4.72 (0.13)	188.8
	pathfinder	4.45 (0.11)	177.8
	pathfinder-thorough	4.35 (0.08)	174.0
$n=40, d=0.7$	minorminer	5.15 (0.06)	206.0
	minorminer-layout	5.23 (0.17)	209.2
	pathfinder	4.95 (0.06)	198.0
	pathfinder-thorough	4.87 (0.03)	194.8

Table 5. Wall-clock after optimization (mean seconds per embedding, clean P_6). “prelim.” is the preliminary single-seed engine (pathfinder-base). Optimized pathfinder matches minorminer to within $\sim 1.3\times$ and is $\sim 3\times$ faster than the preliminary engine; optimized pathfinder-thorough (best-of-4) costs about what a single preliminary run did.

cell	mm	prelim.	pathf.	pf-thor.
$n=30, d=0.3$	0.22	0.94	0.27	1.08
$n=30, d=0.5$	0.70	2.08	0.75	2.31
$n=30, d=0.7$	0.83	2.82	1.03	3.41
$n=40, d=0.3$	0.53	2.12	0.68	2.40
$n=40, d=0.5$	1.15	3.86	1.33	5.08
$n=40, d=0.7$	1.33	5.59	1.87	8.41

within $\sim 1.3\times$ compiled minorminer’s wall-clock—so the chain-length and variance gains come at little time cost. The primitives are classical; the synthesis, application, optimization, and empirical validation are the contribution.

A COMPLETE PER-CELL RESULTS

Table 6 gives the average chain length, mean and standard deviation over the five seeds, for every (source, target, algorithm) in the sweep—the single source of truth for the aggregate statistics in §5. Every cell reaches 100% success. Source families are ER (Erdős-Rényi), reg (d -regular), and BA (Barabási-Albert); the d -regular and BA sources are run on clean Pegasus only. pathfinder-thorough attains the lowest ACL in all 35 cells, and pathfinder never exceeds minorminer.

Table 6. Average chain length, mean(std) over 5 seeds, for all 35 instance-topology cells. Columns: minorminer, minorminer-layout, pathfinder, pathfinder-thorough.

cell	mm	mm-lay	pathf	pf-thor
<i>Pegasus P_6 (clean)</i>				
ER $n=20, d=0.3$	1.79 (0.09)	1.76 (0.05)	1.79 (0.09)	1.65 (0.03)
ER $n=20, d=0.5$	2.28 (0.12)	2.34 (0.06)	2.25 (0.07)	2.19 (0.04)
ER $n=20, d=0.7$	2.59 (0.08)	2.55 (0.09)	2.57 (0.09)	2.51 (0.07)
ER $n=30, d=0.3$	2.70 (0.11)	2.75 (0.15)	2.69 (0.10)	2.59 (0.03)
ER $n=30, d=0.5$	3.30 (0.16)	3.43 (0.10)	3.25 (0.15)	3.19 (0.10)
ER $n=30, d=0.7$	3.83 (0.22)	3.80 (0.17)	3.79 (0.23)	3.65 (0.06)
ER $n=40, d=0.3$	3.56 (0.05)	3.62 (0.14)	3.52 (0.06)	3.41 (0.06)
ER $n=40, d=0.5$	4.57 (0.10)	4.72 (0.13)	4.49 (0.14)	4.38 (0.08)
ER $n=40, d=0.7$	5.15 (0.06)	5.23 (0.17)	5.07 (0.04)	4.97 (0.04)
reg $n=30, k=4$	1.37 (0.06)	1.41 (0.11)	1.37 (0.06)	1.31 (0.03)
reg $n=30, k=6$	1.92 (0.11)	1.94 (0.04)	1.92 (0.11)	1.86 (0.07)
reg $n=40, k=4$	1.51 (0.12)	1.48 (0.06)	1.51 (0.11)	1.35 (0.04)
reg $n=40, k=6$	2.10 (0.13)	2.13 (0.23)	2.09 (0.12)	1.95 (0.03)
BA $n=30, m=3$	1.56 (0.11)	1.51 (0.07)	1.56 (0.11)	1.50 (0.06)
BA $n=30, m=5$	2.22 (0.09)	2.25 (0.12)	2.20 (0.09)	2.09 (0.07)
BA $n=40, m=3$	1.80 (0.06)	1.73 (0.08)	1.79 (0.06)	1.71 (0.09)
BA $n=40, m=5$	2.56 (0.10)	2.63 (0.13)	2.50 (0.08)	2.44 (0.03)
<i>broken Pegasus P_6 (5% faults)</i>				
ER $n=20, d=0.3$	1.85 (0.12)	1.90 (0.15)	1.84 (0.12)	1.70 (0.03)
ER $n=20, d=0.5$	2.26 (0.07)	2.35 (0.10)	2.25 (0.06)	2.23 (0.05)
ER $n=20, d=0.7$	2.73 (0.14)	2.69 (0.16)	2.72 (0.14)	2.56 (0.09)
ER $n=30, d=0.3$	2.79 (0.12)	2.79 (0.12)	2.79 (0.13)	2.67 (0.07)
ER $n=30, d=0.5$	3.53 (0.11)	3.50 (0.10)	3.48 (0.13)	3.34 (0.09)
ER $n=30, d=0.7$	3.85 (0.17)	4.01 (0.16)	3.79 (0.14)	3.68 (0.13)
ER $n=40, d=0.3$	3.62 (0.15)	3.63 (0.19)	3.58 (0.10)	3.52 (0.09)
ER $n=40, d=0.5$	4.88 (0.18)	4.71 (0.19)	4.80 (0.18)	4.52 (0.06)
ER $n=40, d=0.7$	5.52 (0.19)	5.36 (0.20)	5.34 (0.18)	5.08 (0.18)
<i>Zephyr Z_4</i>				
ER $n=20, d=0.3$	1.53 (0.02)	1.55 (0.04)	1.53 (0.02)	1.45 (0.03)
ER $n=20, d=0.5$	1.96 (0.04)	1.94 (0.07)	1.96 (0.04)	1.83 (0.02)
ER $n=20, d=0.7$	2.16 (0.07)	2.35 (0.16)	2.16 (0.07)	2.11 (0.04)
ER $n=30, d=0.3$	2.24 (0.04)	2.28 (0.08)	2.23 (0.06)	2.18 (0.09)
ER $n=30, d=0.5$	2.75 (0.07)	2.73 (0.02)	2.71 (0.07)	2.69 (0.06)
ER $n=30, d=0.7$	3.03 (0.06)	3.03 (0.04)	3.01 (0.05)	2.96 (0.03)
ER $n=40, d=0.3$	2.92 (0.09)	3.11 (0.13)	2.88 (0.09)	2.83 (0.11)
ER $n=40, d=0.5$	3.63 (0.08)	3.60 (0.16)	3.56 (0.07)	3.52 (0.03)
ER $n=40, d=0.7$	4.18 (0.30)	4.29 (0.06)	4.09 (0.26)	3.85 (0.07)

REFERENCES

- [1] David E. Bernal, Kyle E. C. Booth, Raouf Dridi, Hedayat Alghassi, Sridhar Tayur, and Davide Venturelli. 2020. Integer Programming Techniques for Minor-Embedding in Quantum Annealers. In *Integration of Constraint Programming, Artificial Intelligence, and Operations Research (CPAIOR)*. arXiv:1912.08314.
- [2] Kelly Boothby, Paul Bunyk, Jack Raymond, and Aidan Roy. 2020. Next-Generation Topology of D-Wave Quantum Processors. arXiv:2003.00133 [quant-ph] arXiv:2003.00133.
- [3] Kelly Boothby, Andrew D. King, and Jack Raymond. 2021. *Zephyr Topology of D-Wave Quantum Processors*. Technical Report. D-Wave Systems.
- [4] Tomas Boothby, Andrew D. King, and Aidan Roy. 2016. Fast Clique Minor Generation in Chimera Qubit Connectivity Graphs. *Quantum Information Processing* 15 (2016), 495–508.
- [5] Jun Cai, William G. Macready, and Aidan Roy. 2014. A Practical Heuristic for Finding Graph Minors. arXiv:1406.2741 [quant-ph] arXiv:1406.2741.
- [6] Raouf Dridi, Hedayat Alghassi, and Sridhar Tayur. 2018. A Novel Algebraic Geometry Compiling Framework for Adiabatic Quantum Computations. arXiv:1810.01440 [quant-ph] arXiv:1810.01440.
- [7] Pablo Gómez-Tejedor, Eneko Osaba, and Esther Villar-Rodriguez. 2025. Addressing the Minor-Embedding Problem in Quantum Annealing and Evaluating State-of-the-Art Algorithm Performance. arXiv:2504.13376 [quant-ph] arXiv:2504.13376.
- [8] Timothy D. Goodrich, Blair D. Sullivan, and Travis S. Humble. 2018. Optimizing Adiabatic Quantum Program Compilation Using a Graph-Theoretic Framework. *Quantum Information Processing* 17 (2018).
- [9] Andrew Lucas. 2014. Ising Formulations of Many NP Problems. *Frontiers in Physics* 2 (2014), 5.
- [10] Larry McMurchie and Carl Ebeling. 1995. PathFinder: A Negotiation-Based Performance-Driven Router for FPGAs. In *ACM International Symposium on Field-Programmable Gate Arrays (FPGA)*. 111–117.
- [11] Kurt Mehlhorn. 1988. A Faster Approximation Algorithm for the Steiner Problem in Graphs. *Inform. Process. Lett.* 27, 3 (1988), 125–128.
- [12] Riccardo Nembrini, Maurizio Ferrari Dacrema, and Paolo Cremonesi. 2025. Minor Embedding for Quantum Annealing with Reinforcement Learning. arXiv:2507.16004 [quant-ph] arXiv:2507.16004.
- [13] Hoang M. Ngo, Tu N. Do, Duc Vu, Tamer Kahveci, and My T. Thai. 2024. CHARME: A Chain-Based Reinforcement Learning Approach for the Minor Embedding Problem. arXiv:2406.07124 [quant-ph] arXiv:2406.07124.
- [14] Elijah Pelofske. 2024. 4-Clique Network Minor Embedding for Quantum Annealers. *Physical Review Applied* 21 (2024), 034023. arXiv:2301.08807.
- [15] José P. Pinilla and Steven J. E. Wilton. 2019. Layout-Aware Embedding for Quantum Annealing Processors. In *ISC High Performance*.
- [16] Stefano Sinno, Thomas Groß, Nicholas Chancellor, et al. 2025. Optimised Quantum Embedding via Complete Bipartite Graphs. arXiv:2504.21112 [quant-ph] arXiv:2504.21112.
- [17] Yuya Sugie, Yuki Yoshida, Nozomu Mukai, Hiroshi Banno, Hiroyuki Shioda, et al. 2018. Graph Minors from Simulated Annealing for Annealing Machines with Sparse Connectivity. In *Theory and Practice of Natural Computing (TPNC)*.
- [18] Hiromitsu Takahashi and Akira Matsuyama. 1980. An Approximate Solution for the Steiner Problem in Graphs. In *Mathematica Japonica*, Vol. 24. 573–577.
- [19] Cédric Vincent-Cuaz, Rémi Flamary, Marco Corneli, Titouan Vayer, and Nicolas Courty. 2022. Semi-Relaxed Gromov-Wasserstein Divergence and Applications on Graphs. In *International Conference on Learning Representations (ICLR)*.
- [20] Hongteng Xu, Dixin Luo, Hongyuan Zha, and Lawrence Carin. 2019. Gromov-Wasserstein Learning for Graph Matching and Node Embedding. In *International Conference on Machine Learning (ICML)*. arXiv:1901.06003.
- [21] Stefanie Zbinden, Andreas Bärtschi, Hristo Djidjev, and Stephan Eidenbenz. 2020. Embedding Algorithms for Quantum Annealers with Chimera and Pegasus Connection Topologies. In *ISC High Performance*.